

Sunya Afrasiabi

email | personal website | linkedin

Education

University of Pennsylvania, BSE: Mechanical Engineering, Double Major: Philosophy Sept 2024 - May 2028

- **Relevant Coursework:** Thermal-Fluids Engineering, Statics & Strength of Materials, Thermodynamics, Dynamics, Linear Algebra, Differential Equations, Probability, Python, Machining, Epistemology, Social Contract

University at Buffalo, Gifted Math Program (Dual Enrollment) Sept 2023 - April 2024

- **Relevant Coursework:** College Calculus III, Introduction to Linear Algebra

Experience & Leadership

Munch Industries, Inc., Mechanical Engineer July 2026 - Present

- Contribute to mechanical development for Orble, an automated boba tea vending startup, spanning material selection, structural analysis, CAD, and fabrication as part of a small engineering team.
- Determine material and structural specifications through stress/strain analysis, then translate validated designs into fabricated hardware via 3D printing, PVC fabrication, and machining under limited resources and tight timelines.

University of Pennsylvania – Li Group, Undergraduate Researcher May 2026 - Present

- Research insect-wing-inspired morphing structures, modeling compliant geometries in **Rhino** and **Grasshopper** to study parametric mechanical behavior, with the goal of developing a lightweight, composite material for aerospace and soft robotics applications.
- Optimize designs for powder-bed TPU 3D printing at Cornell University, considering printability and structural integrity across hollow vein geometries and wall thicknesses

NASA L'SPACE Proposal Writing & Evaluation Experience, Radiation Subteam Lead Sept 2025 - Dec 2025

- Developed a solution addressing radiation-resistant material limitations in HDPE composites to improve performance under lunar radiation conditions.
- Led radiation research for proposal and sat on a NASA review panel; ranked **top 6 of 40+ national teams**.

#MyStory Program, Founder and Project Manager Jan 2021 - Aug 2024

- Founded and managed a non-profit oral history initiative promoting cultural awareness through exhibitions.
- Secured over **\$24,000** via donations, fundraisers, and grants (Riley's Way Foundation, Prudential, Ashoka, Peter & Elizabeth Tower Foundation).
- Directed operations across **20+ community partners** and **40+ volunteers**.

Technical Projects

CompOS: AI-Powered Component Sourcing Platform Fall 2026

- Built a full-stack application on **Palantir Foundry** integrating **AIP** to evaluate dual-use substitutes for military-grade components as part of a 4-person hackathon team, joining 41 target components with 1,000+ candidate records pulled live from the DigiKey and Nexar/Octopart APIs.
- Designed a custom multi-dimensional scoring algorithm to rank substitution risk and qualification burden, then shipped an AI-guided dashboard that cut a multi-week manual sourcing workflow down to minutes.

Multi-Phase Butane Rocket – Python Numerical Simulation Spring 2026

- Wrote a Python Euler-integration solver coupling butane vaporization, pressure-driven propulsion, drag, and ballistic dynamics to model a variable-mass water rocket in real time.
- Combined the simulation with experimental trajectory reconstruction and aerodynamic/thermodynamic modeling to hit constrained flight paths and minimize target error.

Mars Rover Chassis Prototype Spring 2026

- Designed and optimized a Mars rover chassis prototype for University Rover Challenge (URC) constraints as part of a 3-person engineering team, developing detailed SolidWorks assemblies and structural architectures.
- Engineered multi-axis suspension linkages using Grade 5 titanium rods and 3/8" threaded ball joint sockets, running FEA-driven iterations to optimize load paths and bracket placement under high-impact loading scenarios.

Papers

Linear Algebra Foundations of Neural Networks, University of Pennsylvania August 2025

- Wrote expository research paper connecting linear algebra, neural networks, and robotics. ([Link](#))

Aviation Solutions for Wildfire Prevention, Wharton Undergraduate Aerospace Club May 2025

- Co-authored investor-oriented whitepaper on UAVs, AI wildfire prediction, and aerospace innovations, framing \$6.4B+ market opportunities and potential \$5B annual cost savings. ([Link](#))

Skills

Programming & Data: Python, NumPy, SQL, REST APIs, Numerical Methods, Foundry, AIP, Computational Geometry

Engineering Software: Rhino 3D, Grasshopper, SolidWorks, MATLAB, FEA

Fabrication: Laser Cutting, FDM/SLA 3D Printing, Mechanical Assembly, CNC and Manual Machining

Foreign Languages: English (Native), French (Intermediate)